Appendix E: R for Metrology

Introduction

R (R Core Team 2019) is an open-source programming language and environment that is used for data analysis and statistical computing. R was originally developed in the mid-90s as a free statistical analysis tool influenced by two other programming languages—S and Scheme. It has since grown in popularity, primarily due to the number of R packages that are available for a wide range of applications, including linear and nonlinear regression, machine learning, graphical analysis, and more. There are many advantages to using R, including that it is:

- Open-source and free to download.
- Specifically designed for statistical analysis.
- Easily extended through R packages.
- Supported by a large community of users.
- Excellent for data visualization.

Although there is no official dedicated support for R, there is a group of users known as the Core R Team that maintains and edits the base R source code. There are also many available resources for learning and using R; for more information, see Hornik (2018).

Installation of R

R can be downloaded from the Comprehensive R Archive Network (CRAN) at www.r-project.org (R Core Team 2019). It is available for Windows, Unix-Like, and Mac operating systems with installation instructions provided on the CRAN site. R Studio is a recommended and commonly used (but optional) user interface that can be downloaded from www.rstudio.com (R Studio Team 2015).

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R Packages

An R package is a group of code containing functions, data, etc. that can be used for a specific purpose (e.g., time series analysis). These packages can be developed by individual R users and submitted to CRAN for public release. As mentioned above, R packages extend the functionality of base R by providing code for a spectrum of purposes. Packages can be installed from CRAN via the console in R by using the install.packages() function.

R for Metrology

There are several existing R packages that perform many of the analyses that are used in metrology. Several packages are used throughout this book, such as Deriv for symbolic differentiation, and onls for orthogonal distance regression. NIST has developed a package specifically for metrology called metRology (Guthrie et al. 2015) which performs many of the analyses outlined in the GUM. Pertinent functions in the metRology package include:

- GUM()—Evaluation of uncertainty using the GUM method.
- uncertMC()—Evaluation of uncertainty using the Monte Carlo method.
- welch.satterthwaite()—Calculation of the Welch–Satterthwaite effective degrees of freedom.

The package also includes built-in example data sets and methods for plotting the uncertainty results. However, since the goal of this book was to show the step-by-step calculations used in the GUM, the metRology package was not used.

R can be used in numerous areas of metrology; in this handbook, it was used for the following:

- Exploratory data analysis (Chap. 4).
- Fitting and assessing probability distributions (Chap. 4).
- Numerical integration (Chap. 5).
- Symbolic partial differentiation for the GUM method (Chap. 7).
- Monte Carlo sampling (Chap. 8).
- Performing Analysis of Variance (ANOVA) with both fixed and random effects (Chap. 9).
- Regression (linear, nonlinear, etc.) (Chap. 10).
- Statistical process control (Chap. 11).

Summary

R is an open-source statistical software that can be used for many metrology applications, and its functionality continues to increase with the large number of packages that are developed by R users. While there is a learning curve to programming in R, it provides a powerful tool for creating automated, reproducible analyses using a variety of statistical methodologies. Additional resources for learning R can be found in Wickham and Grolemund (2017) and Teetor (2011).